## 10.2.2. Types of Multimeter

There are two types of multimeters

1. Analog multimeter
2. Digital multimeter

### 10.2.2.1. Analog Multimeter

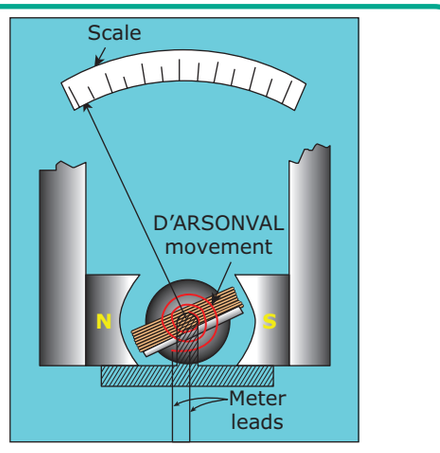
An analog multimeter consists of an ordinary pivoted type of moving coil galvanometer.



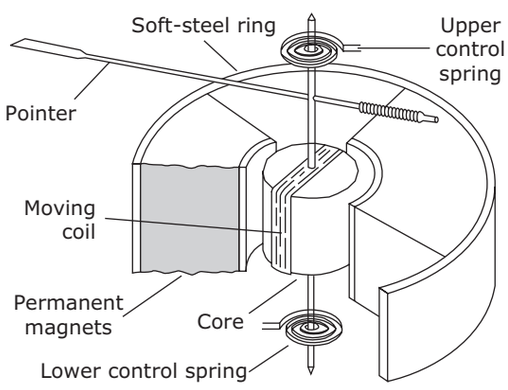

Galvanometer is also called as D'ARSONVAL meter, named after the inventor of galvanometer.

A galvanometer consists of a coil pivoted on a jewel bearing between the poles of a permanent magnet. An indicating needle is fastened to the coil. When electric current passes through the coil, magnetic flux is generated. This magnetic flux is converted into a mechanical force to move the pointer over the scale. The movement of pointer is called as meter reading. After the measurement is finished, the control spring is used to bring back the needle to the resting position (0 or ∞). The analog multimeter components are shown in Figure 10.4.

To measure voltages, currents and resistances, proper circuits are connected with the galvanometer. The galvanometer used in a multimeter is always of left zero type i.e., the needle normally rests in extreme left position, whereas in ordinary galvanometers the needle rests in centre zero position as shown in Figure 10.5.

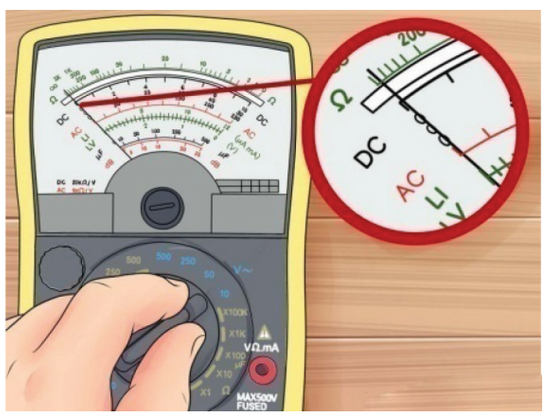



(a) Analog Multimeter Showing Coil

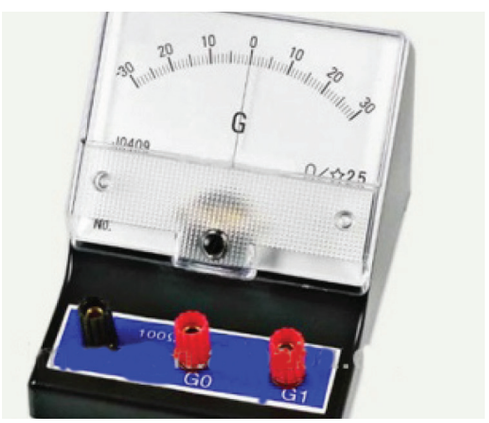



(b) Permanent Magnet and Control Spring

**Figure 10.4** Internal construction of Analog Multimeter




**(a) Multimeter (Left - Zero)**




**(b) Galvanometer (Centre-Zero)**

**Figure 10.5** Needle at Zero Position in Multimeter and Galvanometer



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